

Machine Learning (ML) Projects — Medium-Level Projects

Three practical projects with a short project description and a step-by-step approach.

1. Customer Churn Prediction

Short description: Predict which customers are likely to leave a subscription service so the business can target retention efforts.

How to approach:

- 1 Collect customer demographics, tenure, usage, support interactions, and billing history.
- 2 Handle missing values, categorical variables, outliers, and class imbalance.
- 3 Create features such as usage trend, support-contact frequency, and recent payment changes.
- 4 Train baseline logistic regression and tree-based models such as Random Forest or XGBoost.
- 5 Compare ROC-AUC, precision, recall, F1, and calibration.
- 6 Deploy the best model and create a ranked list of high-risk customers with suggested review reasons.

2. Predictive Maintenance for Industrial Equipment

Short description: Predict whether a machine is likely to fail within a future time window using sensor readings and maintenance history.

How to approach:

- 1 Collect timestamped sensor data such as temperature, vibration, pressure, and operating hours.
- 2 Create rolling statistics and lag features while avoiding future-data leakage.
- 3 Define a failure prediction window and label each observation.
- 4 Train classification models and handle rare failure events with appropriate sampling/weights.
- 5 Evaluate recall, precision, PR-AUC, and the number of false alarms.
- 6 Build a monitoring view showing machine risk scores and trigger alerts for high-risk equipment.

3. Personalized Product Recommendation Model

Short description: Recommend products to users based on their past interactions, purchases, and product similarity.

How to approach:

- 1 Prepare user-item interaction data from views, clicks, carts, and purchases.
- 2 Create a baseline such as popularity-based recommendations.
- 3 Build collaborative filtering or matrix-factorization recommendations.

- 4 Add item metadata to create a content-based fallback for new products.
- 5 Evaluate Precision@K, Recall@K, and NDCG@K using a time-based validation split.
- 6 Create a simple recommendation API and analyze recommendations for different user segments.

Suggested stack: use Python for data preparation and modeling where applicable; Git/GitHub for version control; and a simple Streamlit/API layer when a UI or deployment component is useful.